

YouKnow

Circuit-Modelled Synth for Reason 14+

User Guide - 0.1.0b17 beta



Direct subtractive synthesis, 47 original sounds, a 128-program historical factory bank, polyphonic MIDI, monophonic Note/Gate CV, stereo chorus, and scalable rear-panel engine quality.

Install and update

Install and update YouKnow through your Reason Studios account and the Reason Rack Extension browser. The Universal 45 development archive is for Reason Studios' build service and is not a customer installer.

After installation, find **YouKnow Circuit-Modelled Synth** under Instruments in the device palette. Creating it makes a sequencer track and auto-routes its stereo output when a suitable destination is available.

First sound

1. Create YouKnow and play MIDI notes. The Init patch is loaded by default.
2. Use the patch display and browse controls to choose from 175 supplied patches, including Init. Original sounds are organized into Bass, Leads, Keys, Brass, Pads, Strings, and Effects; the historical programs are in the top-level A and B folders with slot-prefixed names.
3. Follow the front panel from LFO and DCO through HPF, VCF, VCA, ENV, and CHORUS. Performance, keyboard, and unit-character controls occupy the lower row.
4. Save edited sounds with Reason's normal patch command. Engine Quality choices are stored with the song, not with a patch.

Front panel

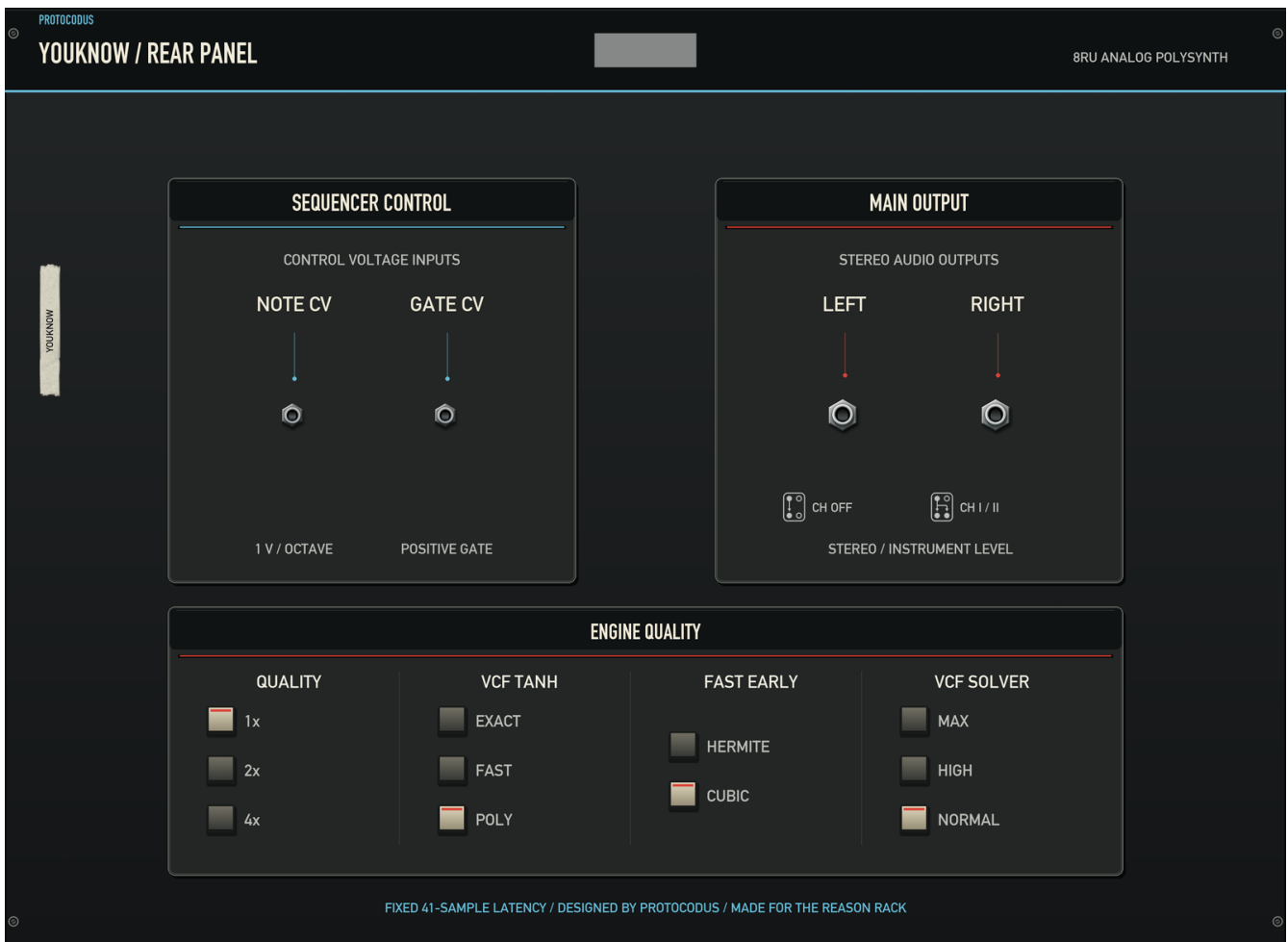
LFO sets modulation rate and delayed onset. **DCO** selects octave range, waveforms, PWM source and depth, sub level, noise, and oscillator modulation. **HPF** provides four high-pass positions including bass boost.

VCF controls cutoff, resonance, envelope polarity and amount, LFO amount, and keyboard tracking. At high resonance the filter can self-oscillate. **VCA** chooses envelope or gate behavior and sets output level. **ENV** is the shared attack, decay, sustain, and release envelope. **CHORUS** offers Off, I, and II.

The lower row contains pitch and modulation wheels, volume, performance modulation depths, glide, key mode, transpose, master tune, velocity response, 1-16 voices, unit character, and chorus hiss.

Clicking the already-selected Poly 1, Poly 2, or Unison button repeats the instrument's held-key reassignment gesture.

Rear panel and engine quality



The rear panel provides monophonic Note and Gate CV inputs, stereo audio outputs, and four song-persistent engine policies. Patch browsing never changes these settings. They remain available to Remote and Reason automation, with the normal Remote and automation indications on the rear controls. The compact front strip is read-only and reports the four current settings. Because these parameters change CPU use or the filter kernel, assign them to hardware controls deliberately.

- **Quality - 1x, 2x, 4x.** New devices use 1x. Higher settings increase the internal rate and CPU cost. A change waits until voices and output tails are quiet, then crosses through a short safety fade. Use 4x for offline or lightly loaded work; 16 voices at 4x is intentionally the heaviest combination.
- **VCF Tanh - Exact, Fast, Poly.** Poly is the efficient default and enables silent-card retirement. Fast uses the previous lookup kernel. Exact restores the continuously rendered reference path and costs substantially more CPU.
- **Fast Early - Hermite, Cubic.** Cubic is the efficient default used with Poly; Hermite retains the earlier interpolation path.
- **VCF Solver - Max, High, Normal.** Normal is the efficient default. High and Max spend progressively more arithmetic on each internal filter sample.

For a fair CPU comparison, create a new device or explicitly select 1x, Poly, Cubic, and Normal. Existing songs correctly retain their saved choices.

Patches

The bank contains 47 original Protocodus patches: Init, 18 featured sounds, and 28 categorized additions spanning bass, leads, keys, brass, pads, strings, and effects. It also contains 128 historical factory tone states in top-level A and B folders, named by their A11-A88 and B11-B88 hardware slots. A patch-only calibration trim keeps browsing levels balanced without altering the tone records or the visible Volume control. Downward attenuation applies immediately when a patch changes; upward makeup ramps briefly to avoid clicks. Reason's metadata and tags make all 175 patches searchable by sound type and character. The full catalog is in `PRESETS.md` in the source distribution.

The Rack Extension does not expose MIDI Program Change or SysEx file import/export; use Reason's patch browser and patch files instead.

CV, tuning, and timing

The rear Note/Gate CV pair is monophonic; MIDI remains polyphonic up to the selected Voices limit. Gate CV triggers a note and Note CV sets its pitch. Changing Note CV while Gate remains high retargets the held note legato. The instrument follows Reason's master tuning preference and reports a fixed 41-sample latency at every Quality setting.

Reason's audio-reset request clears voices, envelopes, chorus state, deferred quality changes, and wrapper note bookkeeping. Transport stop releases sounding notes normally.

Troubleshooting

- **Notes arrive but there is no sound:** load Init or reset the device, verify Volume and VCA Level, and confirm that stereo outputs are connected. Version 0.1.0b17 includes the preset-switch held-key reassignment fix.
- **High CPU:** select 1x, Poly, Cubic, and Normal on the rear. Reduce the Voices setting for dense arrangements. Reserve 4x with many voices for offline or lightly loaded work.
- **A Quality change seems delayed:** release all notes and let the output tail settle; the change is deliberately applied only at a safe silent boundary.
- **No CV response:** connect both Gate and Note CV, then verify the source uses Reason's normal pitch and gate ranges.
- **A problem persists:** include the YouKnow version, Reason version, operating system, computer model, audio sample rate, patch name, and reproduction steps in a message to protocodus@proton.me.

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Audio and control data are processed inside Reason on the user's computer. Reason may store YouKnow settings in songs and patches. Reason Studios handles accounts, licensing, purchases, downloads, host diagnostics, and any related network activity under its own settings and privacy terms.

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Third-party notices and provenance

Protocodus' original YouKnow source and panel artwork are covered by the MIT license. The Rack Extension is built with the separately licensed Reason Rack Extension SDK and is intended for distribution through the authorized Reason Studios channel. Reason SDK material is not relicensed under YouKnow's MIT license.

The pitch and modulation wheels, audio and CV sockets, device-name tape, and back-panel placeholder use the SDK's standard instrument artwork so they behave and read like native Reason controls. The rear mono and spreading symbols are the white variants from Reason Studios' RE2D Stock Graphics 1.1 pack. These Reason assets remain covered by the SDK and stock-graphics terms, not YouKnow's MIT license. Panel text is rasterized during development from Apple-supplied DIN fonts; the font files themselves are not shipped in the Rack Extension.

Modelling references

YouKnow is an independent implementation informed by published virtual-analog research. It does not copy or include the reference implementations associated with these publications.

- Vadim Zavalishin, *The Art of VA Filter Design* - topology-preserving transforms and ladder-filter analysis.
- Tim Stilson and Julius O. Smith, *Analyzing the Moog VCF with Considerations for Digital Implementation* (1996) - ladder root-locus analysis.
- Antti Huovilainen, *Non-linear digital implementation of the Moog ladder filter* (DAFx-04), and Stefano D'Angelo and Vesa Valimaki, *Generalized Moog Ladder Filter: Part II* - nonlinear delay-free-loop methods.
- Vesa Valimaki, Jussi Pekonen and Juhan Nam, *Perceptually informed synthesis of bandlimited classical waveforms using integrated polynomial interpolation* (JASA, 2012) - bandlimited waveform synthesis.
- Martin Holters and Julian Parker, *A Combined Model for a Bucket Brigade Device and its Input and Output Filters* (DAFx-18) - bucket-brigade delay modelling.

Patches and factory tone-memory data

The bank contains 47 patch names and parameter states created for YouKnow by Protocodus. It also includes the historical 128 factory tone states as 2,304 bytes of functional parameter data: 128 records of sixteen 7-bit control bytes plus two packed switch bytes. In physical memory order A11-A88/B11-B88, their SHA-256 is 394ae874da33aa63fa4833932fbf415546d2ad66b1b6b9a36315601799eeec21.

The corpus was mechanically decoded and matched byte for byte across these public archives:

- [Juno-106 Connection](#): Bank A and Bank B tape WAV files and `Factory_Patches.pat`.
- [Jarvik7 Juno-106 Librarian](#): the position-preserving factory .106 library.
- [KR-106](#): its factory PAT and decoded JSON transcription. That repository is published under GPLv3 at the pinned commit; the other two archives state no explicit redistribution license for the tone corpus.

Patch-name text does not exist in the hardware memory records. The committed design data retains archival descriptive labels for provenance; a small number of browser-facing labels are normalized or made commercially neutral. The project's MIT license covers Protocodus' original code and assets and does not assert ownership of Roland trademarks or these historical tone states. It is not established whether the KR-106 repository's GPL declaration governs the functional hardware-memory records, who can license the original factory settings, or whether that license is suitable for an SDK-governed commercial Rack Extension. Do not treat the mirror's GPL

declaration or public download availability as commercial clearance; a qualified rights assessment or direct permission remains required. No data was taken from the separately distributed Roland Cloud Original 128 product.

No firmware, ROM image, sample, impulse response, captured audio, or separately sold sound-bank data is included.

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